

Jesús M. Rueda-Becerril | Ph.D.

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Profile

Computational Scientist with a PhD in Physics and 10+ years developing large-scale physics simulation software, numerical PDE solvers, and scientific ML pipelines for HPC environments. Expertise in multiscale/multiphysics modeling, uncertainty quantification, and synthetic data generation for AI surrogate models. Proven experience with geospatial and remote sensing data processing, parallel computing, and production-grade scientific software development. Seeking to apply physics-based simulation and machine learning to high-impact environmental science problems.

Skills

Languages: Python, C/C++, Fortran 95, Rust, R, Java, SQL, Julia, Shell

HPC & Parallel: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

Simulation: Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs, finite difference/volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

ML & Data: Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, Paraview, VisIt

Experience

Independent Contractor, *Mercor*, Jan 2026 — Present.

- Providing deep expertise in Physics and numerical methods to improve LLM models for top AI labs.

Spatial Data Scientist, *TealWaters*, May 2025 — Jan 2026.

- Led optimization of scientific code for topographic terrain modeling, improving computational performance of elevation derivative calculations
- Engineered Python tools for geospatial data processing and analysis; utilized random forest ML for wetland probability mapping.
- Performed EDA, feature engineering, and geospatial data preparation including multispectral (Sentinel 1/2) processing for ML/AI land-cover models
- Collaborated with software engineers and scientific team to transition prototype simulation tools into production.

Postdoctoral Research Associate, *Rochester Institute of Technology*, Feb 2021 — Apr 2022.

- Led NSF-sponsored project to upgrade PatchworkMHD, a large-scale C-based HPC simulation code for modeling supermassive black hole binaries using state-of-the-art numerical techniques.
- Implemented new physics features (black hole spin) to the simulation library without impacting runtime, demonstrating expertise in large-scale scientific software architecture
- Designed and ran large-scale computational experiments on the Frontera supercomputer (TACC), benchmarking and identifying performance optimization opportunities in parallel HPC code
- Applied scientific machine learning (gradient-based optimization) to calibrate physical model parameters in Paramo against Fermi-LAT telescope observations, demonstrating ML-assisted simulation workflows for complex physical systems
- Collaborated on multi-institutional HPC simulation project studying binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow, *Purdue University*, Oct 2018 – Nov 2020.

- Architected and developed Paramo, an open-source multiscale/multiphysics simulation library in Fortran 95 modeling relativistic particle distributions via the Fokker-Planck equation coupled with radiative transfer across temporal and spatial scales; speedup via OpenMP parallelization; explored GPU acceleration via OpenACC
- Implemented advanced numerical methods including sophisticated PDE solvers, numerical integration, and particle distribution algorithms with arbitrary shape support, without impacting simulation runtime
- Designed Python analysis and visualization tools interfacing with the Paramo simulation library, enabling statistical modeling and parameter optimization workflows
- Obtained and led a NASA grant (\$68K) to model radiation from active galaxies; managed 3 PhD researchers and published 2 peer-reviewed papers

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo*, Jan – Sep 2018.

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios

Software Engineer, *Paychex*, Apr 2022 — Jan 2024.

- Developed Java Kafka consumers for high-throughput data transfer across enterprise databases; deployed to production via Jenkins and OpenShift using CI/CD pipelines
- Conducted SQL data preparation, validation, and analysis from Oracle EBS datasets; built Splunk dashboards for production monitoring
- Developed Python tests ensuring software quality and continuous integration standards

Graduate Research Assistant, *Universitat de València*, Oct 2011 – Jul 2017.

- Independently developed Shell and Python pipelines to run the C-SPEV radiative transfer simulation code and process large HDF5 output datasets
- Implemented numerical tools and data handling extensions to C-SPEV for computing both discrete and continuous spectra from particle distributions with arbitrary shape
- Published 2 first-author papers on numerical simulations of relativistic plasma dynamics

Projects

Blazar Population ML, *Scientific ML (Python, scikit-learn, PyTorch)*, Jan 2026 — Present.

[GitHub](#)

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Forecasting blazar gamma-ray flares using a general-purpose time series foundation model, fine-tuned with Tleco-generated synthetic light curves.

WindsOfChange, *Open-Source Lagrangian Particle Simulation Library (R/C++)*, Aug 2024 — Present.

[GitHub](#)

- Developing C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrains with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco, *Open-Source Physics Simulation Library (Python/Rust)*, Jan 2024 — Sep 2024.

[GitHub](#)

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo, *Open-Source Physics Simulation Code (CONAHCYT Funded)*, Oct 2018 — Apr 2022.

[GitHub](#)

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

PatchworkMHD, *Large-Scale MHD Simulation Code (NSF Project)*, Feb 2021 — Apr 2022.

- Led upgrade of production C-based MHD simulation code for HPC environments; added physics features while maintaining performance at Frontera supercomputer scale.

Education

Ph.D. in Physics, *Universitat de València, Valencia, Spain*, Oct. 2011 – Jul. 2017.

Excellent *Cum Laude*

M.Sc. in Physics, *Universidad Michoacana de San Nicolás de Hidalgo, Morelia, Mexico*, Aug. 2009 – Sep. 2011.

B.Sc. in Physics, *Universidad Autónoma del Estado de México, Toluca, Mexico*, Aug. 2004 – Dec. 2008.

Publications

- [1] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C., in prep.
- [2] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C., in prep.
- [3] **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [4] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, **ApJ** **976**, 182, (2024), [arXiv:2405.17581](#).
- [5] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, **MNRAS** **513**, 5766–5779, (2022), [arXiv:2201.07790](#).
- [6] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, **Phys. Rev. D** **106**, 083015, (2022), [arXiv:2112.09817](#)
- [7] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, **ApJ** **919**, 95, (2021), [arXiv:2106.05356](#)
- [8] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, **MNRAS** **501**, 4092–4102, (2021), [arXiv:2009.02273](#).
- [9] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, **MNRAS** **496**, 974–986, (2020), [arXiv:1910.14049](#).
- [10] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, **MNRAS** **468**, 1169–1182, (2017), [arXiv:1612.06383](#).
- [11] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, **MNRAS** **438**, 1856–1869 (2014), [arXiv:1310.5441](#).
- [12] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, **Phys. Rev. D** **80**, 084023 (2009), [arXiv:1009.1250](#).